

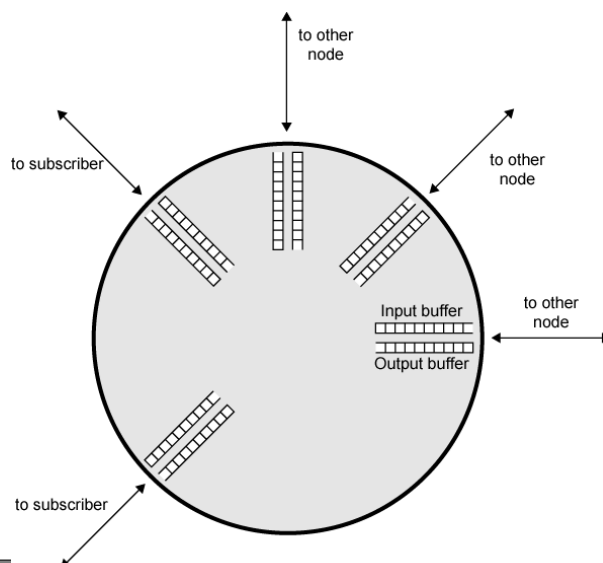
05 The Network Layer

05.06 Congestion Control

What Is Congestion?

- Congestion occurs when the number of packets being transmitted through the network approaches the packet handling capacity of the network
- Congestion control aims to keep number of packets below level at which performance falls off dramatically
- Data network is a network of queues
- Generally 80% utilization is critical
- Finite queues mean data may be lost

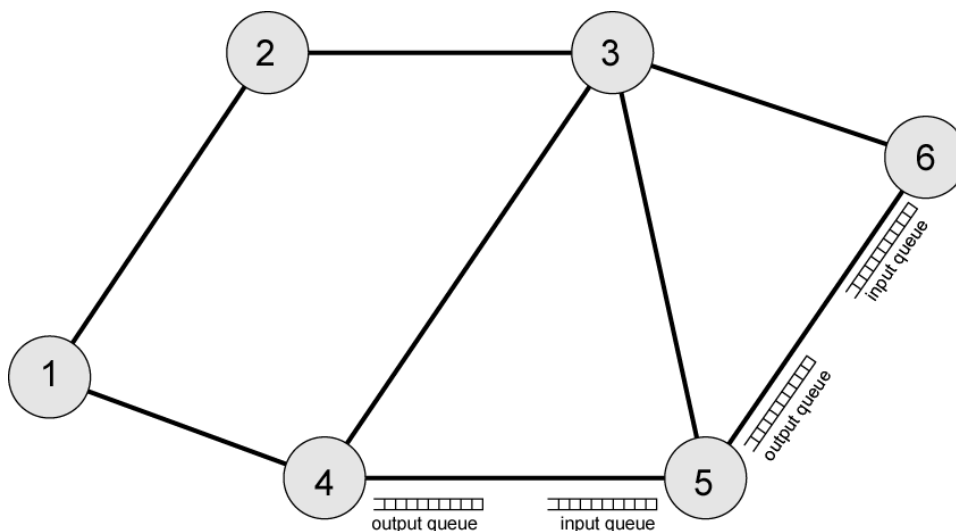
Queues at a Node



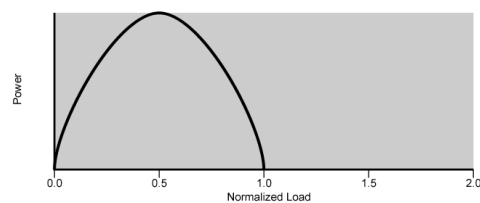
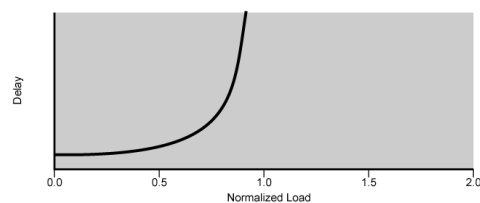
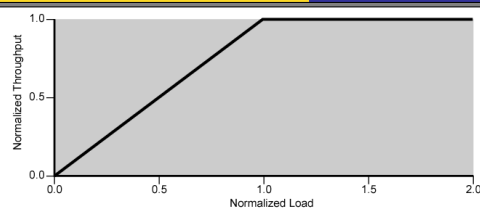
Effects of Congestion

- Packets arriving are stored at input buffers
- Routing decision made
- Packet moves to output buffer
- Packets queued for output transmitted as fast as possible
 - Statistical time division multiplexing
- If packets arrive too fast to be routed, or to be output, buffers will fill
- Can discard packets
- Can use flow control
 - Can propagate congestion through network

Interaction of Queues



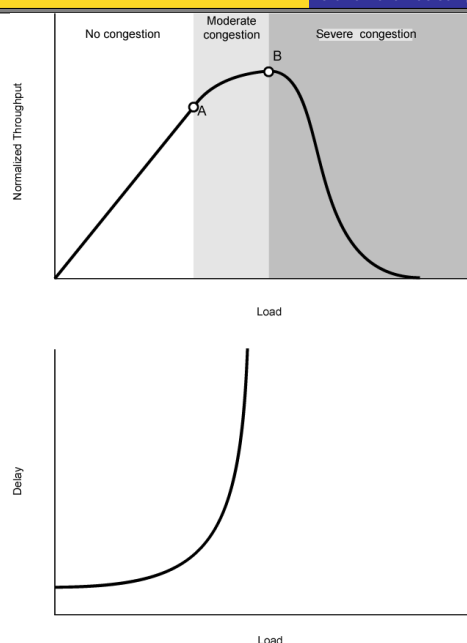
Ideal Network Utilization



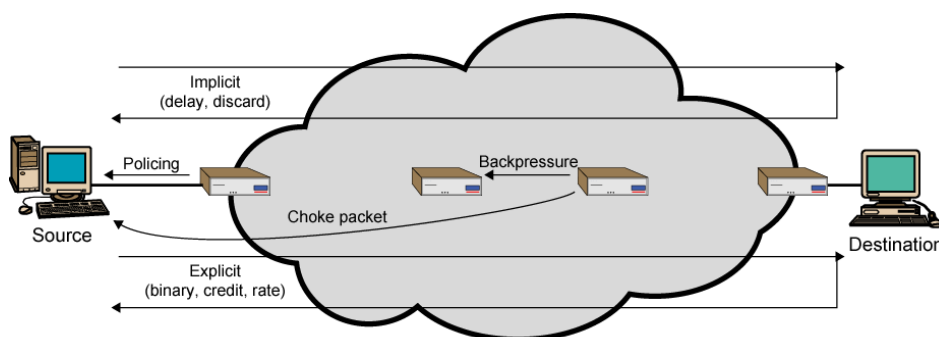
Practical Performance

- Ideal assumes infinite buffers and no overhead
- Buffers are finite
- Overheads occur in exchanging congestion control messages

Effects of Congestion - No Control



Mechanisms for Congestion Control



Implicit Congestion Signaling

- Transmission delay may increase with congestion
- Packet may be discarded
- Source can detect these as implicit indications of congestion
- Useful on connectionless (datagram) networks
 - e.g. IP based
 - (TCP includes congestion and flow control - see transport layer)

Explicit Congestion Signaling

- Network alerts end systems of increasing congestion
- End systems take steps to reduce offered load
- Backwards
 - Congestion avoidance in opposite direction to packet required
- Forwards
 - Congestion avoidance in same direction as packet required

Categories of Explicit Signaling

- Binary
 - A bit set in a packet indicates congestion
- Credit based
 - Indicates how many packets source may send
 - Common for end to end flow control
- Rate based
 - Supply explicit data rate limit
 - e.g. ATM

Traffic Management

- Fairness
- Quality of service
 - May want different treatment for different connections
- Reservations
 - e.g. ATM
 - Traffic contract between user and network

Congestion Control in Packet Switched Networks

- Send control packet to some or all source nodes
 - Requires additional traffic during congestion
- Rely on routing information
 - May react too quickly
- End to end probe packets
 - Adds to overhead
- Add congestion info to packets as they cross nodes
 - Either backwards or forwards

Backpressure

- If node becomes congested it can slow down or halt flow of packets from other nodes
- May mean that other nodes have to apply control on incoming packet rates
- Propagates back to source
- Can restrict to logical connections generating most traffic
- Used in connection oriented that allow hop by hop congestion control (e.g. X.25)
- Not used in ATM nor frame relay

Choke Packet

- Control packet
 - Generated at congested node
 - Sent to source node
 - e.g. ICMP source quench
 - From router or destination
 - Source cuts back until no more source quench message
 - Sent for every discarded packet, or anticipated
- Rather crude mechanism

Choke Packet hop by hop

- Choke packet has a slow reaction
- Control packet
 - Generated at congested node
 - Sent to the previous node
 - Take effect at every hop
- Better than choke packet

Dropping Packets

- **Load shedding:** Wine Vs. Milk
 - Wine: drop new packets (keep old); good for file transfer
 - Milk: drop old packets (keep new); good for multimedia
- **Random Early Detection**
 - When the average queue length exceeds a threshold, packets are picked at random from the queue and discarded.

Random Early Detection (RED)

- Notification is implicit
 - just drop the packet (TCP will timeout)
 - could make explicit by marking the packet
- Early random drop
 - rather than wait for queue to become full, drop each arriving packet with some *drop probability* whenever the queue length exceeds some *drop level*

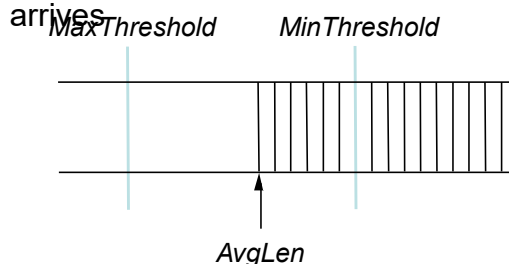
RED Details

- Compute average queue length

$$\text{AvgLen} = (1 - \text{Weight}) * \text{AvgLen} + \text{Weight} * \text{SampleLen}$$

$0 < \text{Weight} < 1$ (usually 0.002)

SampleLen is queue length each time a packet arrives



RED Details (cont)

- Two queue length thresholds

```

if AvgLen <= MinThreshold then
    enqueue the packet
if MinThreshold < AvgLen < MaxThreshold then
    calculate probability P
    drop arriving packet with probability P
if MaxThreshold <= AvgLen then
    drop arriving packet

```

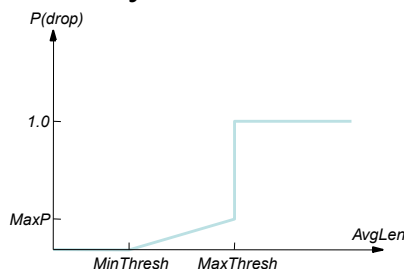
RED Details (cont)

- Computing probability P

$$\text{TempP} = \text{MaxP} * (\text{AvgLen} - \text{MinThreshold}) / (\text{MaxThreshold} - \text{MinThreshold})$$

$$P = \text{TempP} / (1 - \text{count} * \text{TempP})$$

- Drop Probability Curve



Tuning RED

- Probability of dropping a particular flow's packet(s) is roughly proportional to the share of the bandwidth that flow is currently getting
- **MaxP** is typically set to 0.02, meaning that when the average queue size is halfway between the two thresholds, the gateway drops roughly one out of 50 packets.
- If traffic is bursty, then **MinThreshold** should be sufficiently large to allow link utilization to be maintained at an acceptably high level
- Difference between two thresholds should be larger than the typical increase in the calculated average queue length in one RTT; setting **MaxThreshold** to twice **MinThreshold** is reasonable for traffic on today's Internet

Quality of Service

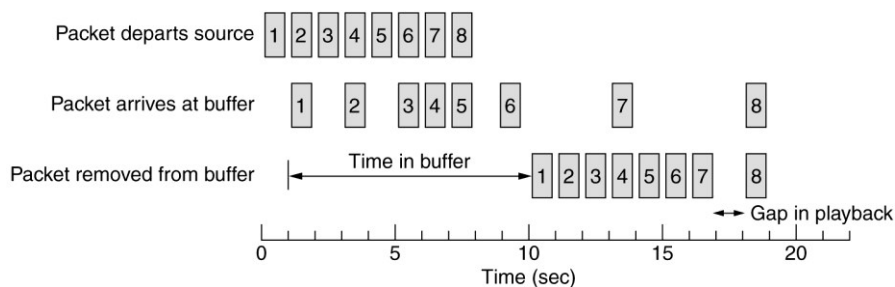
- Requirements
- Techniques for Achieving Good Quality of Service
- Integrated Services
- Differentiated Services

Requirements

Application	Reliability	Delay	Jitter	Bandwidth
E-mail	High	Low	Low	Low
File transfer	High	Low	Low	Medium
Web access	High	Medium	Low	Medium
Remote login	High	Medium	Medium	Low
Audio on demand	Low	Low	High	Medium
Video on demand	Low	Low	High	High
Telephony	Low	High	High	Low
Videoconferencing	Low	High	High	High

How stringent the quality-of-service requirements are.

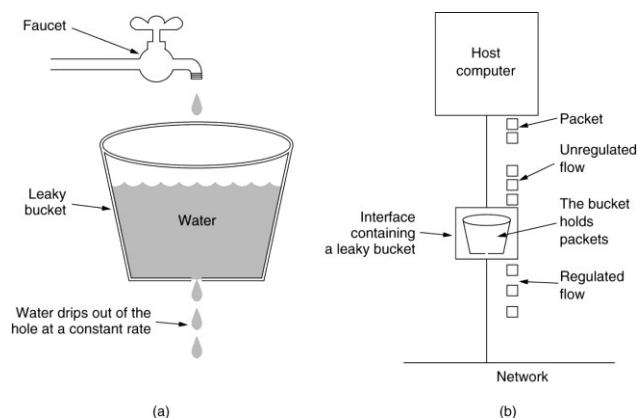
Buffering



Smoothing the output stream by buffering packets.

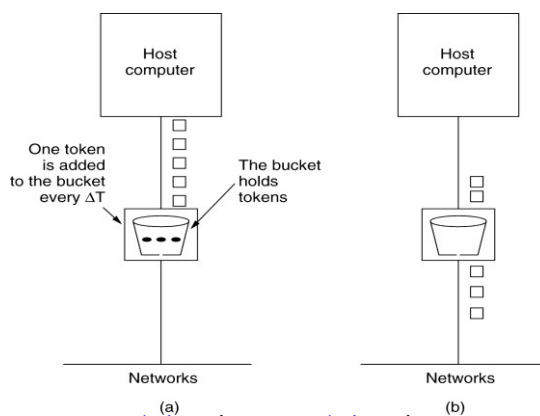
Traffic Shaping

The Leaky Bucket Algorithm



(a) A leaky bucket with water. (b) a leaky bucket with packets.

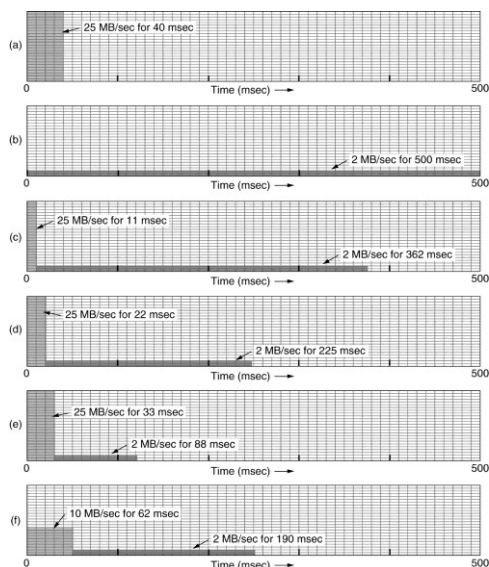
The Token Bucket Algorithm



Token bucket allows some burstiness (up to the number of token the bucket can hold)

The Leaky and Token Bucket Example

- (a) Input to a bucket.
 (b) Output from a leaky bucket.
 Output from a token bucket with capacities of
 (c) 250 KB,
 (d) 500 KB,
 (e) 750 KB,
 (f) Output from a 500KB token bucket feeding a 10-MB/sec leaky bucket.



Resource Reservation

Traffic shaping is more effective when all packets follow the same route.

*We can, similar to virtual circuits, assign a specific route to a flow and then **reserve resources** along that route.*

Three kinds of resources can be reserved:

- Bitrate
- Buffer space
- CPU cycles

Admission Control

We saw, resource reservation but how can the sender specify required resources ? Also, some applications are tolerant of occasional lapses is QoS. Also, apps might not know what its CPU requirements are.

Hence routers must convert a set of specifications to resource requirements and then decide whether to accept or reject the flow.

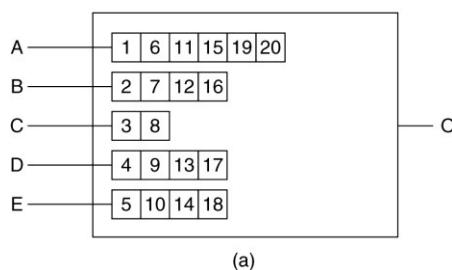
Parameter	Unit
Token bucket rate	Bytes/sec
Token bucket size	Bytes
Peak data rate	Bytes/sec
Minimum packet size	Bytes
Maximum packet size	Bytes

An example of flow specification.

Packet Scheduling

If a router handling multiple flows uses first-come first-served method to process packets, there is possibility of some flows being starved.

- Fair queuing
- Weighted fair queuing



Packet	Finishing time
C	8
B	16
D	17
E	18
A	20

(b)

- (a) A router with five packets queued for line O.
 (b) Finishing times for the five packets.

ATM networks classify flows in four broad categories wrt their QoS demand:

1. *Constant bit rate (e.g., telephony)*
2. *Real-time variable bit rate (e.g., video conferencing)*
3. *Non-real-time variable bit rate (e.g., video streaming)*
4. *Available bit rate (e.g., file transfer)*

Integrated Services (IntServ)

A flow-based approach to QoS using resource reservation.

*Set of protocols aimed at **streaming multimedia** and standardized by the IETF.*

Allows both unicast and multicast transmissions.

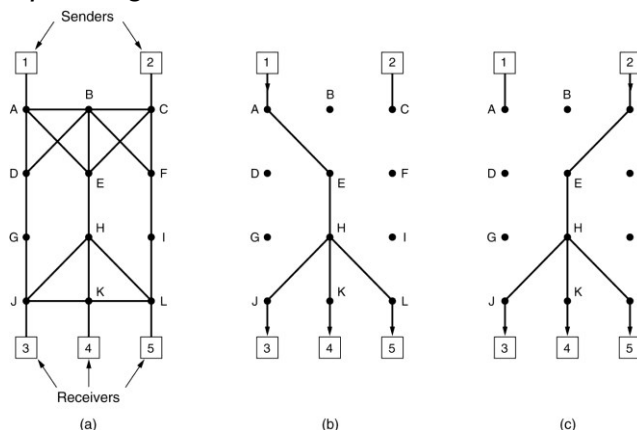
***Resource reSerVation Protocol (RSVP)** is used to reserve the resources at intermediate routers between sender and receivers.*

RSVP allows:

- *Multiple senders to transmit to multiple groups of receivers*
- *Permits individual users to switch channels freely*
- *Optimises bandwidth utilization while simultaneously eliminating congestion.*

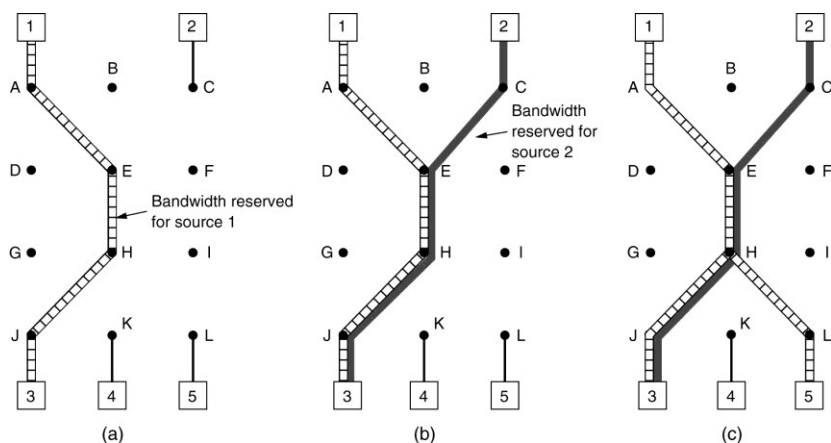
RSVP-The ReSeRvAtion Protocol

Bandwidth reservation is done with reverse path forwarding along the spanning tree.



(a) A network, (b) The multicast spanning tree for host 1.
(c) The multicast spanning tree for host 2.

RSVP-The ReSeRvAtion Protocol (2)



(a) Host 3 requests a channel to host 1. (b) Host 3 then requests a second channel, to host 2. (c) Host 5 requests a channel to host 1.

A lighter approach to QoS

IntServ is very powerful but has some severe drawbacks:

- *There is a setup phase, this causes delay in starting data flow.*
- *Routers need to maintain **per-flow state**. This approach is **flow-based** and not very scalable.*
- *Complex router-to-router exchange of flow information.*

A simpler and approach was then designed by the IETF called, Differentiated Services (DiffServ).

*DiffServ takes a **class-based** (as opposed to IntServ flow-based) approach to QoS*

Differentiated Services (DiffServ)

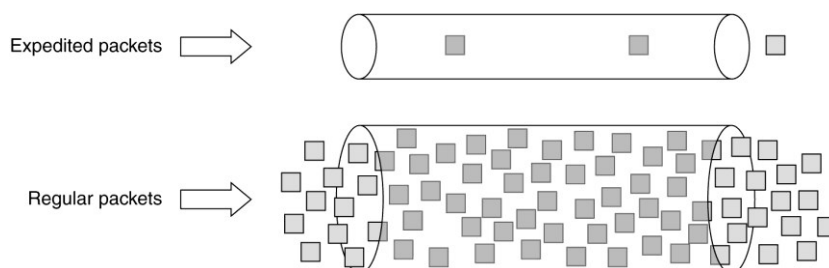
*Introduces **services classes** with corresponding **forwarding rules**.*

*Network operator can “sell” services. Every incoming packet carries a **Type of Service** field. Depending on the service class of a packet, it may receive preferential treatment. The number of classes are decided by the network operator.*

Idea similar to overnight, two-day and surface delivery in courier services.

Two simple classes are: Regular and expedited.

Expedited Forwarding



*Expedited packets experience a traffic-free network,
e.g., if 10% of the traffic is expedited and 90% regular,
20% bandwidth is dedicated to expedited traffic.*

Assured Forwarding

A possible implementation of the data flow for assured forwarding.

There are 4 priority classes and 3 discard probabilities:
low, medium, high.

